TWO EXPLORATIONS AFTER PART III: THE EXACT NORM BY THE SCALLOPED PARSEVAL FORMULA, AND THE EISENSTEIN INTEGERS

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Abstract. Working notes for two questions raised after Part III. (1) The norm $\|G_{1,p}\|^2$ of the cusp form attached to $\theta^2(z)\theta^2(pz)$ is given exactly by a Parseval sum over the cusps whose bottom is the scalloped curve $y=\sqrt{1-\{x\}^2}$ of Waibel's fundamental domain, not the line $y=1$; the formula is evaluated to $p\approx 4\cdot 10^5$, reproduces the Rankin–Selberg values where these have converged, and splits the coefficient B of $\log p$ in $\|G_{1,p}\|^2=\frac{1}{\pi}(\log p)^2+B\log p+O(1)$ into its parts, two of which are determined analytically: 0.383 from the free part of the two main cusps (Part II, Lemma 6.10) and $2\kappa/\pi=0.0288$ from the region between the arcs and the line $y=1$, $\kappa=\int_{-1/2}^{1/2}\log(1-x^2)^{-1/2}\mathrm{d}x$. (2) For the Eisenstein integers, $\theta_3(z)=\sum e((x^2+xy+y^2)z)\in M_1(\Gamma_0(3),\chi_{-3})$ and $\Theta_{1,p}^{(3)}=\theta_3(z)\theta_3(pz)\in M_2(\Gamma_0(3p))$: the level is square-free, all cusps are Atkin–Lehner, the Siegel main term has the same local densities at 3 and at p, the Atkin–Lehner signs are $\Theta|W_p=\chi(p)\Theta$, $\Theta|W_3=-\chi(p)\Theta$, so that the newforms of level $3p$ in the support are exactly those with $a_\varphi(3)=\chi_{-3}(p)$, $a_\varphi(p)=-\chi_{-3}(p)$ (half of them, no inert vanishing), the Rankin–Selberg formula holds with the constant $9\sqrt{3}/(2\pi^2)$ in place of $4/\pi^2$, and $\|G_{1,p}^{(3)}\|^2\sim\frac{3}{2\pi}(\log p)^2$.

1. The exact norm: Parseval over the scalloped domain

Throughout, $G\in S_2(\Gamma_0(L))$, and for a class of pairs (a,c) modulo L with $(c,L)=d$ let $\varrho=\varrho_{a,c}\in\mathrm{SL}_2(\mathbb{Z})$ with first column (a,c), $\hat{d}\in\{d,d/2\}$ as in Part I, Lemma 2.9, $Q=L/\hat{d}$, and $G|_2\varrho=\sum_{k\geq 1}b(k)e(kz/Q)$. Part I, Lemma 2.8 (Waibel's (22)) is

$$\|G\|^2=\frac{\mu}{\tilde{\mu}}\sum_{c\ (L)}\sum_{\substack{a\ (L)\\(a,c,L)=1}}I(\varrho_{a,c}),\qquad I(\varrho)=\int_{\bigcup_{0\leq t<L}t\mathcal{F}}|G|_2\varrho|^2\,\mathrm{d}x\,\mathrm{d}y, \tag{1}$$

and Part II, Section 6.7 bounds $I(\varrho)$ from below and above by the integrals over the strips $y\geq 1$ and $y\geq\sqrt{3}/2$. The exact region is the *scalloped strip*

$$\bigcup_{0\leq t<L}t\mathcal{F}=\big\{-\tfrac{1}{2}\leq x\leq L-\tfrac{1}{2},\ y\geq s(x)\big\},\qquad s(x)=\sqrt{1-\{x\}^2},$$

$\{x\}$ the distance from x to the nearest integer, and s has period 1.

Proposition 1.1 (Scalloped Parseval)**.** *For $j\in\mathbb{Z}$ and $c>0$ let $\Psi_j(c)=\int_{-1/2}^{1/2}e(-jx)\,e^{-c\sqrt{1-x^2}}\,\mathrm{d}x$ (real, $\Psi_{-j}=\Psi_j$). Then*

$$I(\varrho)=L\sum_{k\geq 1}\sum_{j\geq -k/Q}b(k)\overline{b(k+jQ)}\,\frac{\Psi_j(c_{k,j})}{c_{k,j}},\qquad c_{k,j}=\frac{2\pi(2k+jQ)}{Q},$$

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and therefore, with the number $\varphi(L)Q/\,[\hat{d}=d/2\colon 4]$ *of pairs in the class and* $\mu/\tilde{\mu}=1/(L\varphi(L))$,

$$\|G\|^2=\sum_{d\mid L}w_d\sum_{k,j}b_d(k)\overline{b_d(k+jQ_d)}\frac{\Psi_j(c_{k,j})}{c_{k,j}},\qquad w_d=\begin{cases}Q_d & \hat{d}=d,\\ Q_d/4 & \hat{d}=d/2.\end{cases}$$

The terms $j=0$ *with* $\Psi_0(c)$ *replaced by* e^{-c} *are the height-*1 *sums* $\Lambda_1(d)$ *of Part II;* $\Psi_0(c)-e^{-c}>0$ *integrates over the region between the arcs and the line* $y=1$*; the terms* $j\neq 0$ *are correlations* $b(k)\overline{b(k+jQ)}$ *of coefficients at distance a multiple of the width, which a flat bottom annihilates. Since* $c_{k,j}\geq 2\pi j$ *and* $\Psi_j(c)\leq e^{-c\sqrt{3}/2}$*, only* $j\leq 7$ *contribute above* 10^{-16}*.*

Proof. $|G|_2\varrho|^2=\sum_{k,k'}b(k)\overline{b(k')}e((k-k')x/Q)e^{-2\pi(k+k')y/Q}$. Integrating in y from $s(x)$ to ∞ gives $e^{-c\,s(x)}/c$ with $c=2\pi(k+k')/Q$, and since s has period 1 and the x-range consists of L consecutive unit periods,

$$\int_{-1/2}^{L-1/2}e\Big(\frac{(k-k')x}{Q}\Big)e^{-c\,s(x)}\mathrm{d}x=\Big(\sum_{t=0}^{L-1}e\Big(\frac{(k-k')t}{Q}\Big)\Big)\int_{-1/2}^{1/2}e\Big(\frac{(k-k')x}{Q}\Big)e^{-c\,s(x)}\mathrm{d}x,$$

where the geometric sum is L if $Q\mid k-k'$ and 0 otherwise ($Q\mid L$). Writing $k'=k+jQ$ the inner integral is $\Psi_{-j}(c)=\Psi_j(c)$. The count of pairs and $\mu/\tilde{\mu}$ are as in Part II, Lemmas 6.9 and 6.12. □

For $G=G_{1,p}$, $L=4p$, the six classes are $d\in\{1,4,2\}$ (widths $4p,p,4p$; $b=a_{G_{1,p}}/Q$ at the Atkin–Lehner cusps, $b=\frac{1}{p}(T^{\mathrm{odd}}-M^{\mathrm{odd}})$ at the cusp of denominator 2) and $d\in\{p,4p,2p\}$ (widths $4,1,4$). The script `subleading_log_anatomy.py` evaluates the formula with $k\leq 14p$ and $j\leq 8$, Ψ_j by Gauss–Legendre quadrature on a logarithmic grid in c with cubic splines. Checks: $\Lambda_1\leq\|G\|^2\leq\Lambda_{\sqrt{3}/2}$, and agreement with the Rankin–Selberg estimates of Part II where these have converged (7.4286 against 7.4287 at $p=101$).

1.1. **The arc term by Mellin inversion.** At a main cusp of width $Q\in\{p,4p\}$ the free part of the $j=0$ sum is $\frac{1}{Q}\sum_k r(k)^2\Psi_0(4\pi k/Q)/(4\pi k/Q)=\frac{1}{4\pi}\sum_k\frac{r(k)^2}{k}\Psi_0(k/Y)$, $Y=Q/4\pi$. Since

$$\int_0^\infty\Psi_0(c)c^{w-1}\mathrm{d}c=\Gamma(w)\int_{-1/2}^{1/2}(1-x^2)^{-w/2}\mathrm{d}x=\Gamma(w)\big(1+\kappa w+O(w^2)\big),$$

with $\kappa=\int_{-1/2}^{1/2}\log(1-x^2)^{-1/2}\mathrm{d}x=0.045231$, the computation of Part II, Lemma 6.10 ($D(1+w)=4w^{-2}(1+a_1w+\dots)$, residue at $w=0$) gives

$$\sum_k\frac{r(k)^2}{k}\Psi_0(k/Y)=2(\log Y)^2+(c_1+4\kappa)\log Y+O(1),$$

i.e. the arcs add $4\kappa\log Y$ at each main cusp and $\frac{2\kappa}{\pi}\log p=0.028795\log p$ in total. (The first-order approximation $\Psi_0(c)-e^{-c}\approx c\kappa_1e^{-c}$ with $\kappa_1=\int(1-\sqrt{1-x^2})\mathrm{d}x=0.04339$ gives $4\kappa_1$ instead; the higher orders in c also contribute to the $\log Y$ term, and the Mellin computation is exact. The value $\int_{-1/2}^{1/2}\log(1-x^2)^{-1/2}\mathrm{d}x=1-\pi/2+\log 2=0.1223$ proposed for this constant is incorrect; the antiderivative $-\frac{1}{2}(x\log(1-x^2)-2x+\log\frac{1+x}{1-x})$ gives 0.045231.)

1.2. **The anatomy of B.** Write $\|G_{1,p}\|^2-\frac{1}{\pi}(\log p)^2=C_1'+C_2+C_3+C_4$ with $C_1=\frac{1}{4\pi}(S(p/4\pi)+S(p/\pi))$ the free part of the two main cusps, $C_1'=C_1-\frac{1}{\pi}(\log p)^2=0.38312\log p+O(1)$ (Lemma 6.10), $C_2=\Lambda_1(\text{main cusps})-C_1$ (the correction $r(k)^2\to|a(k)|^2$: the Siegel main term for $k<p$, the terms with $\beta\geq 1$ for $k\geq p$), $C_3=\Lambda_1$ of the other four cusps, and $C_4=\|G\|^2-\Lambda_1$

(the part below height 1, split into the diagonal arc terms and the correlations). The numerical results are in Section 3.

2. The Eisenstein integers

Let $\chi = \chi_{-3}$, $N(x,y) = x^2 + xy + y^2$, $r_3(n) = \#\{N = n\} = 6\sum_{d|n}\chi(d)$, $\theta_3(z) = \sum_n r_3(n)e(nz) \in M_1(\Gamma_0(3),\chi)$ (the space is one-dimensional, so θ_3 is the Eisenstein series $E_1(z,\chi)$ with constant term 1), and for a prime $p \neq 3$

$$\Theta_{1,p} = \theta_3(z)\theta_3(pz) = E + G \in M_2(\Gamma_0(3p)), \qquad T(k) = \sum_{\alpha+p\beta=k} r_3(\alpha)r_3(\beta).$$

The level $3p$ is square-free: $\Gamma_0(3p)$ has four cusps, all of them Atkin–Lehner (W_1, W_3, W_p, W_{3p}, widths $3p, p, 3, 1$), and there is no cusp of denominator 2δ.

Proposition 2.1 (Main term). *$E = \sum_k M(k)e(kz)$ with $M(k) = \frac{4\pi^2 k}{3p}\prod_\ell \beta_\ell(k)$, where $\beta_\ell(k) = (1-\ell^{-2})\sigma_{-1}(\ell^{v_\ell(k)})$ for $\ell \nmid 3p$, and, with $v = v_\ell(k)$,*

$$\beta_p(k) = (1+\chi(p)) - \chi(p)\Big(1+\frac{1}{p}\Big)p^{-v}, \qquad \beta_3(k) = (1+\varepsilon) - \varepsilon\cdot\frac{4}{3}\,3^{-v}, \quad \varepsilon = -\chi(p) = \Big(\frac{-p}{3}\Big).$$

Proof. Siegel's theorem for the quaternary form $Q = N(x,y) + pN(z,w)$ of determinant $9p^2$ (Gram matrix) gives $r(\operatorname{gen} Q, k) = 4\pi^2 k\prod_\ell \beta_\ell(k)/\sqrt{9p^2}$ with the usual densities $\beta_\ell(k) = \lim \ell^{-3a}\#\{Q \equiv k \bmod \ell^a\}$, and the genus theta series is the Eisenstein part. At $\ell \nmid 3p$ the discriminant is a square and the form is unimodular (also at $\ell = 2$: $N \oplus N \cong H \oplus H$ over $\mathbb{Z}_2$, the Arf invariants adding to 0), whence the standard formula. At $\ell = p$: the solutions with $(x,y) \not\equiv 0 \pmod p$ are non-singular, lift with multiplicity p^{a-1}, and number $p - \chi(p)$ modulo p for $p \nmid k$ and $(p-1)(1+\chi(p))$ for $p \mid k$; those with $x \equiv y \equiv 0$ reduce, after dividing by p, to the same form with the blocks interchanged and k/p. Hence $\beta_p(k) = 1 - \chi(p)/p$ for $p \nmid k$ and $\beta_p(k) = (1+\chi(p))(1-1/p) + \beta_p(k/p)/p$ for $p \mid k$, which iterates to the closed form. At $\ell = 3$: $N = (x+y/2)^2 + \frac{3}{4}y^2 \cong u^2 + 3v^2$ over $\mathbb{Z}_3$, so $Q \cong u^2 + ps^2 + 3(v^2 + pt^2)$; the solutions with $(u,s) \not\equiv 0 \pmod 3$ are non-singular and number $3-\varepsilon$ for $3 \nmid k$ and $2(1+\varepsilon)$ for $3 \mid k$ modulo 3, with $\varepsilon = (-p/3)$; those with $u \equiv s \equiv 0$ reduce to $3(u'^2 + ps'^2) + (v^2 + pt^2) \equiv k/3$, the same form, with the factor $9/3^3 = 1/3$. So $\beta_3(k) = 1 - \varepsilon/3$ for $3 \nmid k$ and $\beta_3(k) = \frac{2}{3}(1+\varepsilon) + \frac{1}{3}\beta_3(k/3)$ otherwise. The script checks both against direct counts modulo 3^5, p^3 (relative error 10^{-16}), and the whole formula at the level 6, where there are no cusp forms and $T = M$ to 10^{-9} for $k \le 2^{16}$, and at the level 15, where $T - M = \frac{9}{2}f_{15}$. □

Proposition 2.2 (Atkin–Lehner signs and the spectral support). *$\Theta_{1,p}|_2 W_p = \chi(p)\Theta_{1,p}$, $\Theta_{1,p}|_2 W_3 = -\chi(p)\Theta_{1,p}$, $\Theta_{1,p}|_2 W_{3p} = -\Theta_{1,p}$. Consequently $\langle \Theta_{1,p}, \varphi\rangle \neq 0$ for a newform φ of level $3p$ requires $\eta_p(\varphi) = \chi(p)$ and $\eta_3(\varphi) = -\chi(p)$, i.e.*

$$a_\varphi(3) = \chi_{-3}(p), \qquad a_\varphi(p) = -\chi_{-3}(p),$$

and the root number of φ is $+1$.

Proof. For $W_p = \left(\begin{smallmatrix} pa & b \\ 3pc & pd \end{smallmatrix}\right)$, $pad - 3bc = 1$: $W_p = \gamma B_p$ with $\gamma = \left(\begin{smallmatrix} a & b \\ 3c & pd \end{smallmatrix}\right) \in \Gamma_0(3)$, so $\theta_3|_1 W_p = \chi(pd)p^{1/2}\theta_3(pz)$; and $B_p W_p = p\left(\begin{smallmatrix} pa & b \\ 3c & d \end{smallmatrix}\right)$ with the second factor in $\Gamma_0(3)$, so $\theta_3(p\,\cdot)|_1 W_p = p^{-1/2}\chi(d)\theta_3$. The product is $\chi(p)\Theta_{1,p}$. For $W_3 = \left(\begin{smallmatrix} 3a & b \\ 3pc & 3d \end{smallmatrix}\right)$, $3ad - pbc = 1$: both W_3 and $\left(\begin{smallmatrix} 3a & pb \\ 3c & 3d \end{smallmatrix}\right)$ (from $B_p W_3 = \left(\begin{smallmatrix} 3a & pb \\ 3c & 3d \end{smallmatrix}\right) B_p$) are Atkin–Lehner matrices of level 3, and every such matrix is $\left(\begin{smallmatrix} 0 & -1 \\ 3 & 0 \end{smallmatrix}\right)\gamma'$ with $\gamma' = \left(\begin{smallmatrix} c' & d' \\ -3a' & -b' \end{smallmatrix}\right) \in \Gamma_0(3)$; from $\theta_3(-1/(3z)) = -i\sqrt{3}z\,\theta_3(z)$ one gets $\theta_3|_1 W = -i\chi(-b')\theta_3$, b' the upper right entry. The two upper right entries are b and pb, so $\Theta|_2 W_3 = (-i)^2\chi(-b)\chi(-pb)\Theta = -\chi(p)\Theta$, and $W_{3p} = W_3 W_p$ modulo $\Gamma_0(3p)$ gives $-\Theta$. The

script verifies the three signs numerically at a base point (to 10^{-9}). The conditions on φ follow from $\langle\Theta,\varphi\rangle = \langle\Theta|W,\varphi|W\rangle$ as in Part II, Remark 6.8(i), and $a_\varphi(\ell) = -\eta_\ell(\varphi)$ for $\ell \parallel 3p$; the root number is $-\eta_3\eta_p = +1$. □

In contrast with $D = 4$, where every newform of level $4n$ has $\eta_4 = -1$ (Part II, Proposition 6.17) so that the support is empty when $\chi_{-4}(n) = -1$, at the ramified prime 3 both signs of η_3 occur among the newforms of level $3p$ (Steinberg and its unramified twist): for every p the support is the half of the newforms with $a_\varphi(3) = \chi_{-3}(p)$, further restricted by $L(1,\varphi)L(1,\varphi\otimes\chi) \neq 0$.

Proposition 2.3 (Rankin–Selberg)**.** *For a newform φ of weight 2 and level $3p$,*

$$\langle\Theta_{1,p},\varphi\rangle = \frac{9\sqrt{3}}{2\pi^2}\,\chi(p)\,\eta_p(\varphi)\,L(1,\varphi)\,L(1,\varphi\otimes\chi_{-3}).$$

Proof. The proof of Part II, Theorem 6.7(i) applies verbatim: $\theta_3 = E(z,0)$ for the weight-one Eisenstein series of $\Gamma_0(3)$ with character χ (the space is one-dimensional), the cusps of $\Gamma_0(3p)$ over ∞ are $W_Q\infty$, $Q \mid p$, the unfolding gives $\sum_{Q|p}\chi(Q)\Gamma(s+1)(4\pi)^{-s-1}\sum_j r_3(j)\overline{c_Q((p/Q)j)}((p/Q)j)^{-s-1}$, and the Dirichlet series is $6L(w,\varphi)L(w,\varphi\otimes\chi)/(L(2w-1,\chi)\prod_{\ell|p}(1-\chi(\ell)\ell^{1-2w}))$ (at $\ell = 3$ the degenerate factor $(1-a_\varphi(3)x)^{-1}$ equals $L_3(w,\varphi)L_3(w,\varphi\otimes\chi)$ since $\chi(3) = 0$). At $s = 0$ the constant is $\frac{6}{4\pi L(1,\chi_{-3})} = \frac{6\cdot 3\sqrt{3}}{4\pi^2} = \frac{9\sqrt{3}}{2\pi^2}$, against $\frac{4}{4\pi L(1,\chi_{-4})} = \frac{4}{\pi^2}$ for $D = 4$. The script verifies the formula for the newform components of $G_{1,p}$, $p \le 37$, with ratios 1.000 ± 0.003. □

2.1. **The norm.** With $D_3(s) = \sum r_3(k)^2k^{-s} = 36\,\zeta(s)^2L(s,\chi)^2/((1+3^{-s})\zeta(2s))$ and $L(1,\chi_{-3}) = \pi/(3\sqrt{3})$, $36L(1,\chi)^2\cdot\frac{3}{4}\cdot\frac{6}{\pi^2} = 6$, and the computation of Part II, Lemma 6.10 gives

$$S_3(Y) = \sum_k \frac{r_3(k)^2}{k}e^{-k/Y} = 3(\log Y)^2 + c_1'\log Y + c_2',$$

$$c_1' = 6\Big(\gamma + 2\frac{L'}{L}(1,\chi_{-3}) - 2\frac{\zeta'}{\zeta}(2) + \frac{\log 3}{4}\Big) = 16.370,$$

with $\frac{L'}{L}(1,\chi_{-3}) = \gamma + \log 2 + \log\pi - 3\log\frac{\Gamma(1/3)}{\Gamma(2/3)} = 0.368282$ (from the functional equation and $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log 2\pi$) and $c_2' = 13.06$ numerically. All four cusps carry $G_{1,p}$ itself (Proposition 2.2), so Part II, Lemma 6.9 gives

$$\Lambda_h(G_{1,p}) = \frac{1}{4\pi}\sum_k\frac{|a(k)|^2}{k}\sum_{Q\in\{3p,p,3,1\}}e^{-4\pi hk/Q}, \qquad \Lambda_1 \le \|G_{1,p}\|^2 \le \Lambda_{\sqrt{3}/2},$$

and the two wide cusps give the leading term $\frac{1}{4\pi}(S_3(3p/4\pi) + S_3(p/4\pi)) = \frac{3}{2\pi}(\log p)^2 + 0.7130\log p + O(1)$; so $\|G_{1,p}^{(3)}\|^2 \sim \frac{3}{2\pi}(\log p)^2$, against $\frac{1}{\pi}(\log p)^2$ for $D = 4$ (Part II, Theorem 6.14), the ratio 3/2 being $(6/4)^2\cdot(L(1,\chi_{-3})/L(1,\chi_{-4}))^2\cdot\frac{(1+\frac{1}{2})/(1+\frac{1}{3})}{1}\cdot\frac{\zeta\text{-factors}}{}$, i.e. the ratio of the residues 6/4 of D_3 and D_4. [The proof of the asymptotic formula, Part II, Theorem 6.14, carries over with the three parts of Proposition 6.13 for the four Atkin–Lehner classes and no odd classes; we have not written it out.]

3. Numerical results

3.1. **The anatomy of B (`results/subleading_log_anatomy.txt`).** The exact norm of Proposition 1.1 for twelve primes from 101 to 400009 ($k \le 14p$, $j \le 8$; a few seconds per prime). It lies inside the sandwich at every p, and agrees with the Rankin–Selberg estimate of Part II

where that estimate has converged: 7.42858 against 7.42871 at $p = 101$, 10.1189 against 10.1162 at 211; at $p = 1009$ the estimate (16.937, window 2^{21}) is already 0.018 below the exact value 16.955, and at $p = 10007$ it is 3.3 below. The parts of $\|G_{1,p}\|^2 - \frac{1}{\pi}(\log p)^2$:

p	C_1'	C_2	C_3	C_4 arc, main	free part	C_4 corr.	total, / $\log p$
101	1.128	−0.815	0.154	0.164	0.151	−0.001	0.649, 0.141
1009	1.999	−0.711	0.162	0.259	0.218	−0.002	1.727, 0.250
10007	2.876	−0.582	0.166	0.352	0.284	−0.003	2.830, 0.307
100003	3.758	−0.453	0.169	0.446	0.350	−0.004	3.939, 0.342
400009	4.289	−0.374	0.171	0.502	0.390	−0.005	4.607, 0.357

(C_3 includes the arc terms of the four narrow cusps, 0.02; the correlation terms of the narrow cusps are below 10^{-4}.) Least-squares fits $b \log p + c$ over the six primes $p \geq 10^4$:

part	b	c	analytic
C_1' (free part, main cusps)	0.38292	−0.651	$b = (c_1 - 2\log 4\pi^2)/2\pi = 0.38301$
C_2 (Siegel term, $k \geq p$)	0.05656	−1.104	
C_3 (other four cusps, height 1)	0.00138	0.153	
C_4 arc, main cusps	0.04074	−0.023	free part: $b = 2\kappa/\pi = 0.02879$ (measured 0.02879)
C_4 arc, other cusps	0.00057	0.017	
C_4 correlations ($j \neq 0$)	−0.00046	0.001	
$\|G_{1,p}\|^2 - \frac{1}{\pi}(\log p)^2$	0.48170	−1.607	$a = 0.31845$ in the three-parameter fit ($1/\pi = 0.31831$)

So $B \approx 0.48$, not 0.46: Remark 6.16 of Part II fitted Rankin–Selberg estimates that had not converged. Of the 0.48, 0.383 is the free part (proved), 0.029 the arcs acting on the free part (proved above), 0.012 the arcs acting on the Siegel and tail corrections, 0.057 those corrections themselves at height 1, and 0.002 the other cusps; the correlation terms $a(k)a(k + jQ)$ are negligible ($-0.0005 \log p$). The part below height 1 is $0.041 \log p$, against the bound $0.134 \log p$ of Part II, Section 6.7 and the $0.04 \log p$ observed there. The two remaining pieces with a $\log p$ coefficient, C_2 and the arc term on the corrections, are sums over k of $r(k)M(k)$, $M(k)^2$, $r(k)T'(k)$, $T'(k)^2$ against $e^{-k/Y}$ or Ψ_0; their constants are computable by the same Mellin method from the Dirichlet series of $r\beta$ and β^2, which we have not done.

3.2. **The Eisenstein integers (`results/prime_discriminant_norm.txt`).** Local densities: the closed forms of Proposition 2.1 agree with the counts modulo 3^5 and p^3 ($p = 2, 5, 7, 11$) to 10^{-16}; at the level 6, $T = M$ for all $k \leq 2^{16}$ to 10^{-9}; at the level 15, $T - M = \frac{9}{2} f_{15}$ to 10^{-10}, and f_{15} has $a(3) = -1 = \chi(5)$, $a(5) = +1 = -\chi(5)$. Atkin–Lehner: the three signs of Proposition 2.2 hold to 10^{-15} at $p = 5, 7, 11, 13$. Spectral support: for $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$ the Hecke decomposition of $G_{1,p}$ (by T_2, then T_5, T_7 where eigenvalues coincide) gives 21 newform components of level $3p$ and 8 oldform components from level p; *every* newform component has $a(3) = \chi_{-3}(p)$ and $a(p) = -\chi_{-3}(p)$, for both signs of $\chi_{-3}(p)$ ($p = 7, 13, 19, 31, 37$ with $+1$: $1, 2, 2, 3, 4$ newforms; $p = 5, 11, 17, 23, 29$ with -1: $1, 1, 2, 2, 3$ newforms, and $0, 1, 1, 2, 2$ oldforms, all with $a_f(p) = +1$, i.e. root number $+1$); the shares of the norm sum to 1.000 ± 0.005. Rankin–Selberg: for the 21 newform components, $\langle \Theta, \varphi \rangle / \big(\frac{9\sqrt{3}}{2\pi^2} \chi(p) \eta_p(\varphi) L(1, \varphi) L(1, \varphi \otimes \chi) \big)$ lies in $[0.994, 1.005]$ (the twist has level $9p$ in every case, root numbers $+1$ throughout). The norm:

p	RS (2^{21})	Λ_1	$\Lambda_{\sqrt{3}/2}$	$\frac{3}{2\pi}(\log p)^2$	$\Lambda_1/(\log p)^2$	$\mathrm{RS}/(\log p)^2$
11	3.024	2.888	3.335	2.745	0.502	0.526
101	11.893	11.621	12.513	10.170	0.546	0.558
1009	26.452	25.966	27.300	22.842	0.543	0.553
10007	(42.04)	45.385	47.151	40.510	0.535	
100003		69.921	72.120	63.287	0.528	

(the estimate at $p = 10007$ has not converged, as for $D = 4$), with the fit $\Lambda_1 = 0.4815(\log p)^2 + 0.687 \log p - 1.81$ over $p \geq 100$, against the leading term $3/(2\pi) = 0.4775$ and the main-cusp coefficient 0.713 of $\log p$. $S_3(Y) - 3(\log Y)^2 - c_1' \log Y$ is 13.0585 at $Y = 10^4, 10^5$, confirming $c_1' = 16.370$. The exact norm by Proposition 1.1 applies to $D = 3$ as well (four Atkin–Lehner classes, no odd class) and has not been run.

Scripts. `code/subleading_log_anatomy.py` (imports the tables and the Siegel main term of Part II from `../two-squares-mean-square/code`), `code/prime_discriminant_norm.py` (self-contained). Floating point throughout.